

NAGA VENKATA JAHNAVI CHAKKA

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EDUCATION

State University of New York

Masters of Science in Computer Science | GPA: 3.7

Binghamton, NY

Aug 2023 – May 2025

EXPERIENCE

Watson School of Computing and Applied Sciences

Research Assistant

May 2025 – Present

Binghamton, NY

- Designed backend service of an application to facilitate structured group interactions and shared activities using Java and Spring Boot with a layered architecture .
- Built RESTful APIs to handle user onboarding, profile management, form submissions, and session workflows for scalable data transfers between the systems.
- Managed relational data using PostgreSQL with normalized schemas to ensure data integrity and reliability.
- Developed initial frontend components using React and Tailwind CSS to capture structured user input and align UI behavior with backend API contracts.
- Refined API contracts, database schemas, and service logic through iterative development to accommodate evolving requirements and mitigate integration issues.

Motto Systems

Software Engineer

Aug 2022 – Jul 2023

Hyderabad, India

- Developed Java-based backend applications using Spring Boot to support pharmaceutical client workflows, ensuring scalable service design, data integrity, and backend reliability.
- Transitioned backend services from a monolithic architecture to Spring Boot microservices, leveraging AWS services including ECS, EKS, and CloudFormation to enable faster deployments and improved system stability.
- Streamlined CI/CD pipelines and deployment automation using GitHub Actions, enabling automated builds, testing, and containerized deployments with Docker and Kubernetes to improve release consistency across environments.
- Optimized data processing and storage efficiency by designing PostgreSQL schemas and tuning SQL queries, improving transactional performance and backend responsiveness.
- Monitored backend services in deployments using AWS CloudWatch logs to validate system behavior and stability.
- Collaborated with frontend, QA, and platform teams to define backend interfaces, support integration efforts, and resolve service-level issues during testing and release cycles.

PROJECTS

MedQuery AI

- Built a RAG system for medical queries, utilizing LangGraph for multi-step reasoning and integrating Hugging Face Embeddings and Llama 3 8b for retrieval from ChromaDB.
- Optimized retrieval efficiency by implementing advanced chunking and caching strategies, reducing query latency by 40% and enhancing system responsiveness for real-time medical question answering.

Microservices eCommerce Platform

- Developed Spring Boot microservices ecosystem with AI-driven product recommendation pipelines and scalable data flow.
- Applied distributed systems with PostgreSQL, caching layers, and load balancing for high availability and performance.
- Leveraged Azure App Service for scalable deployment with observability via telemetry dashboards and alerting tools.

Staff Management System

- Developed a web-based staff scheduling system using PHP, JavaScript, and MySQL, reducing manual scheduling errors.
- Implemented role-based access control (RBAC) for administrators and employees to manage workload distribution.
- Optimized database queries and designed responsive interfaces to enhance usability, reliability, and performance under multi-user workloads.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript

Databases: PostgreSQL, MySQL, Redis, Cassandra

Frameworks: Spring Boot, Spring MVC, RESTful Web Services, JPA/Hibernate, Angular, React

Tools: Git, GitHub Actions, Docker, Kubernetes, Linux

Cloud: AWS (EC2, S3, RDS, DynamoDB, IAM, CloudWatch), Azure (App Service, Blob Storage, Azure SQL Database)

Others: Object-Oriented Programming, Layered Architecture, Microservices, API Design, CI/CD Pipelines, Agile

PATENT

Natural Language Processing Patent Grant – QA and IoT-based Smart Hanger System (IPR92781082)